

Dhruv Kohli

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EDUCATION

The University of Chicago (GPA: 3.94/4.0)

Chicago, IL

Master of Science in Financial Mathematics

Expected December 2026

- Courses: Option Pricing, Portfolio Management, Statistical Inference (PhD), Time Series Analysis (PhD)

B.S. in Mathematics with Honors & B.A. in Economics with Honors

June 2025

- Courses: Econometrics, Machine Learning, Multivariate Statistics, Real & Complex Analysis, Probability

TRADING COMPETITIONS & AWARDS

- 15th Annual IAQF Student Competition: Winner – Represented UChicago as Team Leader
- 2025 CME Trading Challenge: 2nd Place worldwide trading commodities – Leader of Team Inefficient Markets
- 2024 UChicago College New Venture Challenge: 3rd Place as COO of Resolv, presenting to UChicago and investors

SKILLS

Technical: Python (NumPy, pandas, XGBoost), SQL, Databricks; machine learning, convex optimization, Monte Carlo

Markets: Options, futures, fixed income, commodities; option pricing, portfolio optimization, hedging; Akuna Options 101/201

EXPERIENCE

MET Sales and Trading

Zug, CH

Market Risk Intern – LNG and Power Desks

June 2026 – Present

- Built a multi-tenor Bachelier style Monte Carlo pricer for clean spark spread options using AR(p)-GARCH(1,1) volatility and skewed/heavy-tailed innovations, generating prices and Greeks across user selected tenors
- Develop Power BI, Databricks, and Python analytics to quantify delta/gamma and market, basis, FX, and volume exposures across power and LNG derivatives, supporting daily hedging decisions and long-term contract risk reviews

Exponential Technology

Chicago, IL

Quantitative Researcher – UCPL Program

October 2025 – March 2026

- Engineered an ensemble machine learning model for Non-Farm Payroll forecasting that reduced 50-month walk-forward root mean squared error by 20.8% versus economist consensus and outperformed consensus in 74% of months

AXA Investment Managers (Acquired by BNP Paribas)

Paris, FR

Assistant Portfolio Manager Intern – Emerging Markets Fixed Income Team

June 2024 – August 2024

- Implemented dynamic Nelson-Siegel models to analyze yield curve sensitivity using cleaned live Bloomberg and proprietary data, identifying maturity buckets with significant z-score deviations for relative value opportunities

Assistant Portfolio Manager Intern – Emerging Markets Fixed Income Team

June 2023 – August 2023

- Built a Bloomberg application to optimize a bond portfolio given a chosen DV01 and yield from an approved universe

Resolv (Web3 Asset Protection Start-Up)

Chicago, IL

Chief Operating Officer

October 2023 – December 2024

- Orchestrated a \$450k pre-seed round and scaled users to 2,000+, securing \$100M in LOIs for platform assets

RESEARCH & PROJECTS

Machine Learning Forecasting Platform for ERCOT Power Markets

April 2026 – July 2026

- Built point-in-time, hub-pooled quantile gradient-boosting models to forecast ERCOT wind, temperature, and load
- Verified 23% / 14% / 2.5% MAE reductions (temperature / wind / load) vs. ERCOT's forecast, with walk-forward CV

Options-Only Portfolio Optimization with Greek-Based Risk Modeling

February 2026 – July 2026

- Adapted Markowitz theory to option-only books via Greek factor covariance and a cost aware convex conic solve
- Walk-forward backtest on 2018-2026: 1.18 Sharpe / 2.28 Sortino net, validated via Monte Carlo simulations and CPCV

University of Chicago Economics Department Thesis

November 2024 – May 2025

- Authored honors thesis estimating heterogeneous effects of Syrian refugee integration on Canadian student outcomes using augmented inverse-propensity weighting and PISA data, with variation by gender and urbanicity

LEADERSHIP & ACTIVITIES

University of Chicago, Economics & Mathematics Departments

Chicago, IL

Teacher's Assistant – Microeconomics, Pre-Calculus, Calculus, Number Theory

September 2022 – March 2026

- Lead discussion sections and office hours, developed and graded problem sets and exams, and ran review sessions

Phoenix Funds

Chicago, IL

Member / Executive Board Member / Co-President

September 2021 – June 2025

- Facilitated learning of fixed income and derivatives for underclassmen focusing on real-world applications

ADDITIONAL INFORMATION

Languages: English, French (Native), Hindi (Heritage Speaker), German (Basic), Mandarin (Basic)

Interests: Basketball, Board Games, Chess (UChicago club board member), Cooking, Poker, Rugby